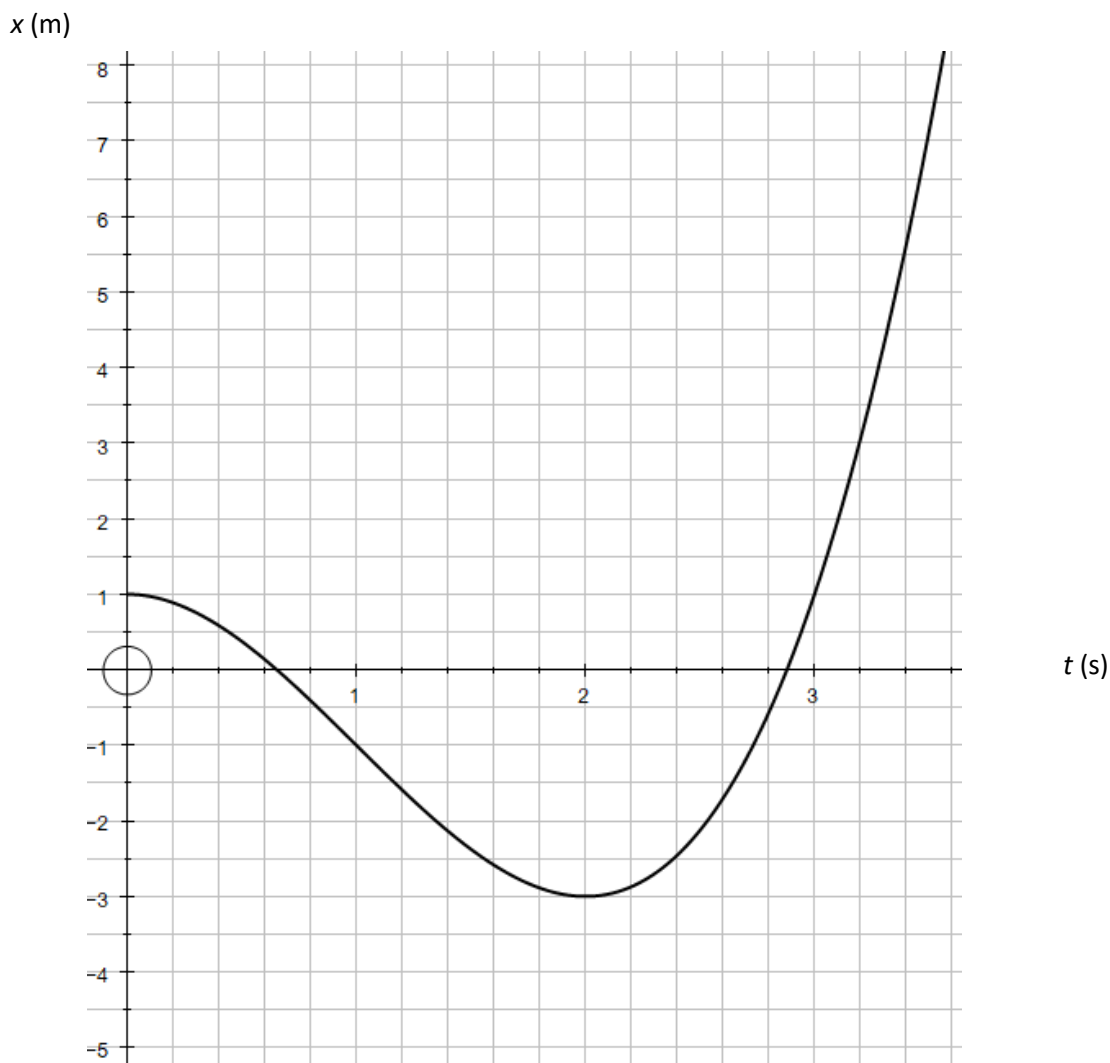


Position, Velocity, Acceleration

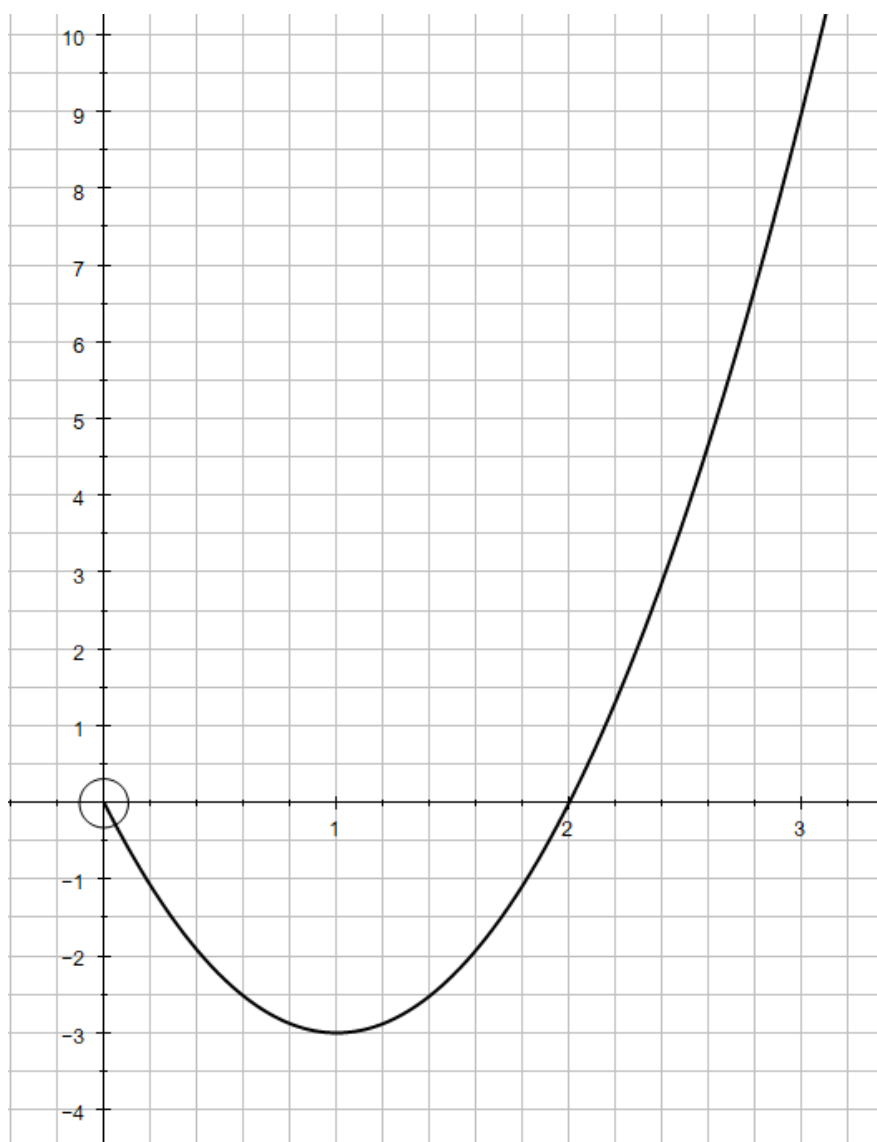
An object is moving along the x -axis. The graph below shows its position, x metres beyond the origin, as a function of the elapsed time, t seconds.



- At what times does the particle pass through the origin?
- Which way is the particle moving when $t = 1$? When $t = 2.5$?
- At what time(s) is the particle stationary?
- The *displacement* over a particular time interval is the change in position over that interval. What is the displacement over the following time intervals?
 - $t = 2$ to $t = 3$
 - $t = 0$ to $t = 3$
 - $t = 0$ to $t = 2$
- What is the average velocity over these time intervals?
 - $t = 2$ to $t = 3$
 - $t = 0$ to $t = 3$
 - $t = 0$ to $t = 2$
- What is the total distance travelled by the particle over the interval $t = 0$ to $t = 2$?
- What is the total distance travelled by the particle over the interval $t = 1$ to $t = 3$?
- By adding suitable tangents to the graph, complete the following table of *instantaneous* velocities as accurately as you can.

Time (s)	0	1	2	3
Position (m)				
Velocity ()				

I used a computer to find the velocity at every instant from $t = 0$ to $t = 3$, and plotted the result on the graph below.



- Add suitable labels and units to the axes.
- The graph cuts the horizontal axis at $t = 2$. Describe the motion of the particle at that time. Note how this feature appears on the position-time graph.
- The graph has a minimum point when $t = 1$. Describe the motion of the particle at that time. How does this feature appear on the position-time graph?
- Acceleration* is the rate of change of velocity. What are the units of acceleration?
- What is the average acceleration over the interval $t = 1$ to $t = 3$?
- Adding suitable tangents, complete the following table of *instantaneous* accelerations.

Time (s)	0	1	2	3
Position (m)				
Velocity ()				
Acceleration ()				

- The acceleration is not constant. Can you find a formula that gives the acceleration as a function of time?
- Harder.** Can you find a formula giving the velocity as a function of time? What about the position?